

5.2.1 Scope of AI in Education in Pakistan

پاکستان کے تعلیمی نظام میں مصنوعی ذہانت آہستہ آہستہ اہم کردار ادا کر رہی ہے، جیسے لرننگ مینجمنٹ سسٹم (LMS)، خو...

CS XI

Chapter 5: Application and Impacts of Computing

Lecture 07 of 12

5.2.1 Scope of AI in Education in Pakistan

LEARN IT

Artificial Intelligence is gradually becoming an important part of Pakistan's education system, improving **teaching**, **learning** and **administration** (managing the school: admissions, records, timetables).

Practical uses:

- **Learning Management Systems (LMS)** — online platforms (e.g. Google Classroom, Moodle) where teachers post lessons and assignments; AI features can recommend lessons to each student.
- **Automated grading and digital testing** — MCQ tests marked instantly by computer.
- **AI-based career counselling** — software that studies a student's marks and interests and suggests suitable fields.

Benefits:

- Helps students learn **at their own speed**.
- **Identifies students who need extra help**.
- Supports students with disabilities (**Assistive AI**), e.g. text-to-speech for a visually impaired student.

Major challenges in Pakistan:

- Limited **internet access in rural areas**.
- Lack of **computers and digital devices** for all students.
- Need for proper **teacher training** to use AI tools.
- **Data privacy and security** issues — students' personal data must be protected.

Practical uses	Benefits	Challenges
LMS	Learn at own speed	Limited rural internet
Automated grading & digital testing	Identifies students needing help	Lack of computers/devices

AI career counselling	Supports students with disabilities	Teacher training needed
—	—	Data privacy & security

WORKED EXAMPLE

Scenario — Fatima in Larkana and an AI tutor app:

Step 1. Fatima takes a 10-question online maths quiz on the college LMS. It is graded instantly (automated grading).

Step 2. The AI notices she got all 3 trigonometry questions wrong.

Step 3. It recommends two short videos and five easy practice questions on trigonometry — she learns at her own speed.

Step 4. Her teacher's dashboard flags "Fatima – needs help in trigonometry" (identifies students who need help).

Challenge: At home Fatima's village has weak 3G signals, so the videos buffer — showing the rural internet challenge.

COMMON MISTAKE

Mistake: Writing that AI will replace teachers. **Fix:** The source says AI *improves* teaching and needs *teacher training* — teachers use AI as a tool; they are not replaced.

اردو میں سمجھیں

پاکستان کے تعلیمی نظام میں مصنوعی ذہانت آہستہ آہستہ اہم کردار ادا کر رہی ہے، جیسے لرننگ مینجمنٹ سسٹم (LMS)، خودکار جانچ اور کیریئر کونسلنگ۔ اس سے طلبہ اپنی رفتار سے سیکھ سکتے ہیں اور کمزور طلبہ کی نشاندہی ہوتی ہے۔ لیکن دیہی علاقوں میں انٹرنیٹ کی کمی، کمپیوٹرز کی قلت، اساتذہ کی تربیت اور ڈیٹا کی حفاظت بڑے چیلنج ہیں۔

CHECK YOURSELF

1. Give three practical uses of AI in education.
2. State any two challenges of AI in education in Pakistan.

Answers: 1. LMS, automated grading and digital testing, AI-based career counselling. 2. Any two: limited rural internet, lack of devices, need for teacher training, data privacy and security.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. AI in Pakistani classrooms can:

- a. Personalise practice, mark MCQs, translate
- b. Replace all teachers tomorrow by law
- c. Lay fibre
- d. Set primary keys

MCQ 2. A risk of AI homework tools is:

- a. Students submit work they cannot explain
- b. Faster bubble sort
- c. Better NAND
- d. Longer OSI

MCQ 3. Adaptive learning means:

- a. Difficulty changes with the student
- b. Everyone gets the same paper always
- c. No data is used
- d. Only analog TV

MCQ 4. Teachers still needed because:

- a. Care, judgment, fairness, lab skill
- b. AI cannot use electricity
- c. Books are illegal
- d. BIEK forbids computers

MCQ 5. Data used by education AI must be:

- a. Protected (student privacy)
- b. Posted publicly with names
- c. Sold always
- d. Ignored

MCQ 6. A realistic near-term use is:

- a. Urdu/English support and drill quizzes
- b. Fully autonomous colleges
- c. Replacing SDLC
- d. Replacing ER diagrams

Extra short questions

1. Two uses of AI in education in Pakistan.
2. Two limits.
3. What should a student still be able to do unaided?
4. One policy idea for a college.

Extra long question

1. Write a balanced note: opportunities of AI for BIEK CS students vs three rules a college should set.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Language practice; auto-marking MCQs; plagiarism hints; accessibility readers.
2. Bias, hallucinations, connectivity gaps, cheating.
3. Explain the answer in the board paper without the chatbot.
4. AI allowed for practice, banned in the exam hall; teach citation.

Long

1. Opp: extra drills, Urdu help, code explainers. Rules: no AI in exams, no pasting private data into public bots, teacher reviews flagged work. Link to digital literacy.

Next lecture: 5.3 Information Sources (Primary, Secondary, Tertiary)